

GAURAV BOSE

KOLKATA, WEST BENGAL, INDIA

+918017469943 simonwwe96@gmail.com [linkedin.com/in/gaurav-bose-a7a199200/](https://www.linkedin.com/in/gaurav-bose-a7a199200/) github.com/DodoIsADud/
<https://dodoisadud.github.io/>

Education

Jadavpur University

Bachelor of Civil(Construction) Engineering, CGPA:-7.5

Dec. 2021 – May 2025

Kolkata, West Bengal

Jodhpur Park Boys' School

WBCHSE Board: 88.2

2018 – 2020

Kolkata, West Bengal

St.Lawrence High School

WBBSE Board: 89.29

Till 2018

Kolkata, West Bengal

Relevant Coursework

- Data Structures and Algorithms
- Machine Learning
- Database Management
- Data Science
- Optimization Techniques
- Artificial Intelligence
- OOPs

Experience

MITACS Globalink - McMaster University, Canada

May 2024 – August 2024

Machine Learning Intern

Hamilton, Ontario

- Developed Machine Learning models (Tree-SHAP, Random Forest) predicting concrete strength with 95-percent accuracy.
- Optimized concrete mix proportions using data-driven insights to enhance compressive and flexural strength, achieving desirable model accuracy ranging over 90-percent.
- Analyzed datasets using TensorFlow and Colab, and reported actionable insights and parameter dependencies.

Research Intern - Jadavpur University

Feb 2023 – May 2023

Machine Learning Intern

Kolkata, West Bengal

- Built a predictive model (RMSE = 5.50) for concrete strength using Support Vector Regression, Random Forest.
- Applied PowerTransformer and GridSearchCV for model tuning with an accuracy of 86.94-percent on test data.
- Did univariate and multivariate analysis with independent and polynomial categorical features.

Projects

Fraud Transaction Detection Analysis | Python, OpenCV, TensorFlow, Jupyter

February 2023

- Trained Transformer-based fraud detection model with 94 Percent precision; KAVACH Hackathon Regional Finalist.
- Trained on KNN classification model (mean precision: -0.94) and Transformers with Synthetic Minority Oversampling Technique (SMOTE) analysis for better performance.

Portfolio Optimization using Stock Market Data | Python, PyTorch

August 2023

- Modelled efficient frontier portfolios using quadratic programming and risk-return tradeoff.
- Derived key metrics (asset weights, expected return, risk) to recommend optimal portfolio allocations based on investor goals and risk tolerance.

Netflix Dashboard | PowerBI, BigQuery

November 2023

- Developed an interactive dashboard visualizing global content trends and user activity, demonstrating a 113-percent annual increase in Netflix user engagement across all streaming platforms.

Licenses And Certifications

Forage Market Virtual Experience by J.P. Morgan, Great Learning Marketing Foundations

Technical Skills

Languages: Python, C++, SQL, Javascript

Developer Tools: VS Code, Google Colab, Jupyter Notebook, Git, Power-BI, MS Office, React.js

Technologies/Frameworks: TensorFlow, PyTorch, PySpark, React

Achievements And Extracurricular

- Coordinator - E-Cell, GDSC Lead at Jadavpur University
- Events Coordinator of F.E.T.S.U. Presents Sanskriti 2025 at Jadavpur University
- Placement Coordinator at Faculty of Engineering and Technology, Jadavpur University
- Finalist - IIM Indore i5 Summit and State Grappling Gold Medalist
- Winner - Hult Prize On-Campus Round 2024 and Boston Regional Finalists (UHPC-205 Concrete Pitch)